AI interaction in sports

GR1 Needfinding T2 Agentic Interaction

Team members

Owen Kang

Computer Science
UCSB, Senior

Seoul, Korea

Yulin Yu

Computer Science
UCSB, Junior

Kaiping, China

Jiemi Niu

Computer Science
UCSB, Senior

Hefei, China

Choosing Domain

Owen Kang

- I want to focusing on sports training because athletes often have their hands and attention occupied.
- It can make users communicate their state and intent with AI more actively.

Yulin Yu

- I was interested in applying AI as an office worker that users can guide, and interrupt.
- This approach allows users to get help based on their goals and understandings.

Jiemi Niu

- Considering both perspectives, I thought sport contexts might work better as a shared domain.
- Since compared to office workflows, sports make the limits of chat more visible and reduce ambiguity around interaction needs.

Final Domain

Agentic interaction in sports training,
where athletes actively shape AI coaching through non-text interactions.

Who to ask?

Participant selection criteria

- Different kind of training methods
  - Recreational?
  - Professional?
- Diversity
  - Gender
  - Culture
  - Location
- Non-UCSB Students

Possible users and Extreme users

- Recreational runners and weight trainers who train consistently or randomly
- Individuals who actively manage and adapt their training
- Extreme user, such as a high frequency or highly intentional athlete

We have participants...

Jackson

Mechanical Engineering
UC Berkeley, Junior

Bay Area, CA

Tom

Art
UC Riverside, Sophomore

Los Angeles, CA

Jenny

Communication,
USC, Graduate Student

Los Angeles, CA

And we want to ask them:

Can you walk me through your most recent training session, from start to finish?

How do you track your training?

How do you want others help you when training?

Where were you training, and what was the goal for that session?

What signals from your body were you paying attention to?

Were there moments where you adjusted your pace, intensity, or plan?

During training, how do you know when you're getting tired and when you want to stop?

What were you thinking at the beginning of the session?

When do you typically look back at your training information?

If you could express your current state during training in any way, what would feel most natural?

Jackson
College Student,
Have been climbing for 2 years
(Extreme User)
Gym interview, 25 min

"I climb for the challenge and the fun. Help only matters when I'm truly unsure."

Interviewer: Owen Kang

Note taker: Owen Kang

Say

I've been climbing about 2 years

I don't want someone telling me every move

I try different body positions through trial and error

Feedback after climbing helps me improve faster

Discovering the beta myself is part of the fun

I learn more when I reflect after the climb

I try to climb regularly as I can

I do bouldering more than lead climbing

I don't want too much distraction during the climb

Watching climbing videos helps a lot

I ask other climbers at the gym for beta sometimes

Think

Finding beta on my own is rewarding

Quick hints are better than long explanations while climbing if needed

Over intervention might reduce enjoyment

Too much guidance during a climb can break focus and creativity

Structured feedback after climbing would accelerate improvement.

Improvement is about efficiency, not just repetition

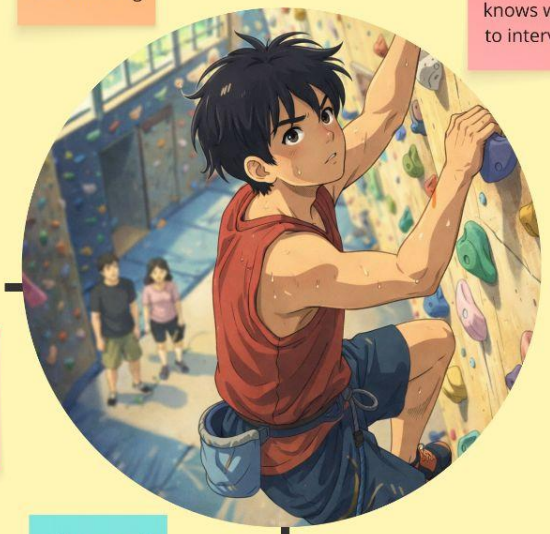
If I fail repeatedly, I want to know what to adjust

AI could be useful if it knows when to intervene

Timing of feedback matters more than amount

Good coaching should adapt to my climbing level

Following how professional climbers do is the fastest way to improve



Do

I usually climb twice a week

Works through problems independently first.

Tries multiple attempts on the same route

Experiments with body positioning and approaches

Asks other climbers for beta when stuck

Reflects after the climb on what didn't work

I film my self climbing to give me a self feedback

Seeks feedback after finishing a climb session

Try not to get scared when trying hard routes

I try to fully focus on the route

I revisit the same route across multiple sessions

Feel

Curious and motivated when facing a challenge

Satisfied when discovering a new beta

Slightly overwhelmed if given too much input in mid climb

Frustrated when unable to finish a problem

Gets really scared on dyno problems

More confident with post climb feedback

AI coaching with motion tracking must be useful

Climbing with peers makes me climb better

Physical strength is more important than skills

Taller the better, still can overcome with trials

Encouragement and coaching might make me improve a lot

Say

I've been climbing about 2 years

I don't want someone telling me every move

I try different body positions through trial and error

Feedback after climbing helps me improve faster

Discovering the beta myself is part of the fun

I enjoy discovering beta on my own and don't want too much guidance during a climb, but AI feedback afterward could help me improve faster

I do bouldering more than lead climbing

I don't want too much distraction during the climb

Watching climbing videos helps a lot

I ask other climbers at the gym for beta sometimes

Think

Finding beta on my own is rewarding

Quick hints are better than long explanations while climbing if needed

Over intervention might reduce enjoyment

Too much guidance during a climb can break focus and creativity

Climbing works best when I stay focused during the attempt, and AI could be useful when it knows when to step back and when to give feedback

Improvement is about efficiency, not just repetition

If I fail repeatedly, I want to know what to adjust

AI could be useful if it knows when to intervene

Timing of feedback matters more than amount

Good coaching should adapt to my climbing level

Following how professional climbers do is the fastest way to improve

Do

I usually climb twice a week

Works through problems independently first.

Tries multiple attempts on the same route

Experiments with body positioning and approaches

I climb regularly, experiment independently, seek help when stuck, and ask others or watch video of myself to review and reflect after climbing

I film my self climbing to give me a self feedback

I revisit the same route across multiple sessions

Seeks feedback after finishing a climb session

Try not to get scared when trying hard routes

I try to fully focus on the route

Feel

Curious and motivated when facing a challenge

Satisfied when discovering a new beta

Slightly overwhelmed if given too much input in mid climb

I feel focused and motivated while climbing, and more confident when understand how to improve afterward

Frustrated when unable to finish a problem

Gets really scared on dyno problems

AI coaching with motion tracking must be useful

Climbing with peers makes me climb better

Physical strength is more important than skills

Taller the better, still can overcome with trials

Encouragement and coaching might make me improve a lot

Insight

Climbers value the challenge and joy of figuring things out themselves, yet still face moments of doubt when they're unsure if a move or body position is correct.

Need

The climber needs AI that quietly supports them by clarifying uncertainty only when it arises, helping them stay confident and immersed in the climb without taking away the joy of exploration.

Tom

College Student,
Relatively new to weight training
Hiking, 20 min

"The only time I felt unsure or stuck was when I did not know how to use a machine, or when I was not sure whether my posture was correct."

Interviewer: Yulin Yu

Note taker: Jiemi Niu

Say

I usually go to the campus gym with my friend

This makes it easy to see progress over time

If it felt unusually easy/difficult/uncomfortable, I make a short note

I followed a routine I saved on my phone

I stop when my form starts to break down

I change an exercise if I feel pain or excessive fatigue

I track weight, reps, and sets

An AI coach might help when I don't know a machine or my posture is correct

I decide based on performance and form

After a workout, I check if I hit my planned weights, reps, sets

I want to ensure progressive overload instead of guesswork

Think

Progress should be intentional and measurable

Near failure is useful, but breakdown is a red flag

External validation is useful when I don't have clarity

I want to avoid injury or bad habits

Machines and posture are areas where I lack certainty

AI could be helpful if it fills specific gaps

Tracking prevents wasted effort and random training

If something felt off, I should capture it for future adjustment

Good form matters more than lifting heavier weight

I don't need constant guidance—only situational help

Do

Uses dumbbells and barbell exercises confidently

Follows a saved routine in a fixed sequence

Swaps exercises when pain/fatigue arises

Goes to the campus gym with a friend

Adjusts load based on successful reps

stop a set when the form starts to break down or when reach near failure

Uses Instagram/friends as reference points for training ideas

Logs sets/reps/weight during rest periods

Review pass workout to assess progress

Adds short notes about difficulty or discomfort

Feel

Slightly uncertain in unfamiliar situations

Reassured by data and consistency

Curious about improvement

Cautious about injury or incorrect posture

Open to support if it reduces uncertainty without disrupting flow

Wants to feel competent and in control

Calm and methodical during workouts

Motivated by progress and improvement

Satisfied when hitting the plan

Open to support but not depending on it

Yulin Yu

Say

I usually go to the campus gym with my friend

This makes it easy to see progress over time

If it felt unusually easy/difficult/uncomfortable, I make a short note

I track weight, reps, and sets

Would want AI help mainly when unsure about machine use or posture

I change an exercise if I feel pain or excessive fatigue

After a workout, I check if I hit my planned weights, reps, sets

An AI coach might help when I don't know a machine or my posture is correct

I decide based on performance and form

I want to ensure progressive overload instead of guesswork

Think

Progress should be intentional and measurable

Near failure is useful, but breakd a rec

External validation is useful when I don't have clarity

I want to avoid injury or bad habits

I mostly self-directed, but I lack certainty with unfamiliar equiment and posture

AI could be helpful if it fills specific gaps

Tracking prevents wasted effort and random training

If something felt off, I should capture it for future adjustment

Good form matters more than lifting heavier weight

I don't need constant guidance—only situational help

Do

Goes to the campus gym with a friend

Uses dumbbe and barb exercise confiden

Reviews sessions afterward and leaves short adjustment notes for next time

Swaps exercises when pain/fatigue arises

Adjusts load based on successful reps

stop a set when the form starts to break down or when reach near failure

Uses Instagram/friends as reference points for training ideas

Logs sets/reps/weight during rest periods

Review pass workout to assess progress

Adds short notes about difficulty or discomfort

Feel

Slightly uncertain in unfamiliar situations

Reassured by data and consistency

Curious about improvement

Cautious about injury or incorrect posture

Open to support if it reduces uncertainty without disrupting flow

Motivated and satisfied when progress is visible

Wants to feel competent and in control

Motivated by progress and improvement

workouts

Satisfied when hitting the plan

Open to support but not depending on it

Yulin Yu

Insight

Even those confident weight trainer who train based on data will encounter uncertain moments, especially when dealing with unfamiliar equipment and movement postures, because relying solely on intuition and self-monitoring is simply not enough at this time.

Need

The user needs on-demand, situational guidance that helps validate form and equipment use only at moments of uncertainty so they can keep training confidently without breaking flow.

Jenny
Graduate Student,
Recreational runner,
Eating in a restaurant, 30 min

"If someone could tell me what to do, that would actually be helpful."

Interviewer: Jiemi Niu

Note taker: Yulin Yu

Say

"I'm a 'recreational' runner, just for some exercise"

"I generally don't run for long on my own"

"It can be really helpful if someone just know how I'm doing"

"I'm afraid of getting hurt because of wrong way of running"

"I'm not that professional"

"When running, I think of random things like homework, dinner, or losing weight"

"If someone could tell me what to do, that would actually be helpful."

"I watch some Tiktok videos about running"

"I don't really look at detailed stats while running"

"Talking while running is exhausting"

"I can tell how I'm doing by how hard I'm breathing"

"I don't want to think too much"

Think

As long as moving her body, that should be good enough

Want to run for relaxing from assignments and losing weights

Body already shows how she's doing

Stats not really matter much for her

Sometimes running is tiring especially not with friends

She don't want to constantly make decisions while running

Only know few about what "good training" looks like

Video/Audio can be great to show people's states

Running should feel simple, not stressful

Body already shows how I'm doing—I don't need to explain it

Talking while running takes too much effort

Do

Starts running without a strict plan

Adjusts speed naturally when breathing becomes heavy

Runs more based on habit than goals

Slows down when she feels overheated or uncomfortable

Rarely talks while running, even with others

Lets others set the pace when running with someone

Don't usually track training stats when/after running

Don't have a professional way of running

Glad to see if someone would like to help/guide

Not run everyday

Feel

Slight uncertainty about whether she's exercising "correctly"

Relaxed when starting a run

Gratitude when help doesn't require her to talk

A little bit lazy to look for a better way

Relief when someone else takes the lead

Satisfaction from simply being active

Curious about right way

Boring if running needs think much

Tried if running on her own

Have no time to make a strict plan

Say

"I'm a 'recreational' runner, just for some exercise"

"I generally don't run for long on r

"It can be really helpful if someone just know how I'm doing"

"I'm afraid of getting hurt because of wrong way of running"

I would like to have someone help me when running, but not via talking

"I'm not that professional"

"When run think of ra things homewo dinner, or weigh

"I watch some Tiktok videos about running"

"I don't really look at detailed stats while running"

"Talking while running is exhausting"

"I can tell how I'm doing by how hard I'm breathing"

"I don't want to think too much"

Think

As long as moving her body, that should be good enough

Want to run for relaxing from assignments and losing weights

Body already shows how she's doing

Stats not really matter much for her

Sometimes running is

She don't want onstantly decisions e running

Only know few about what "good training" looks like

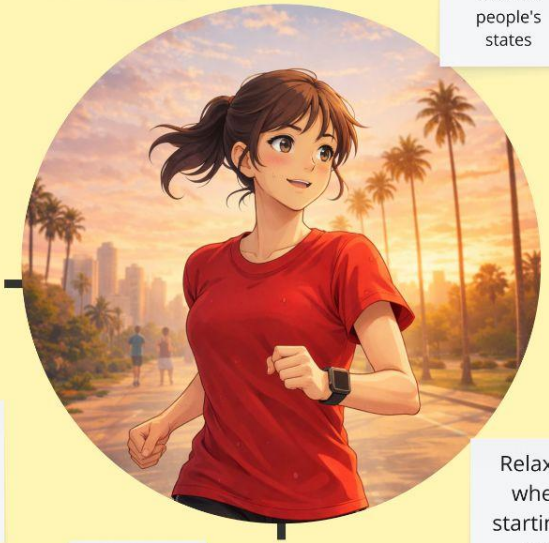
Recreational exercise is enough, and body shows one's state

Video/Audio can be great to show people's states

y already s how I'm g—I don't need to explain it

stressful

Talking while running takes too much effort



Do

Starts running without a strict plan

Adjusts speed naturally when breathing becomes heavy

Runs more based on t than oals

No strict plan is made for running

Slows down when she feels overheated or uncomfortable

hers set e when g with eone

Don't usually track training stats when/after running

Don't have a professional way of running

Glad to see if someone would like to help/guide

Not run everyday

Feel

Slight uncertainty about whether she's exercising "correctly"

Relaxed when starting a run

Gratitude when help doesn't her to k

A little bit lazy to look for a better way

If learning the right way and making proper plans for her can be easy, that's great

Satisfaction from simply being active

Curious about right way

running needs think much

Tried if running on her own

Have no time to make a strict plan

Insight

Recreational runners like Jenny are open to guidance during training, but lack the expertise and energy to explicitly communicate their physical state—especially while in motion.

Need

Recreational runners need support that can understand their physical state through observable cues (e.g., breathing, sweating, posture) and give guidelines based on these cues.

Summary

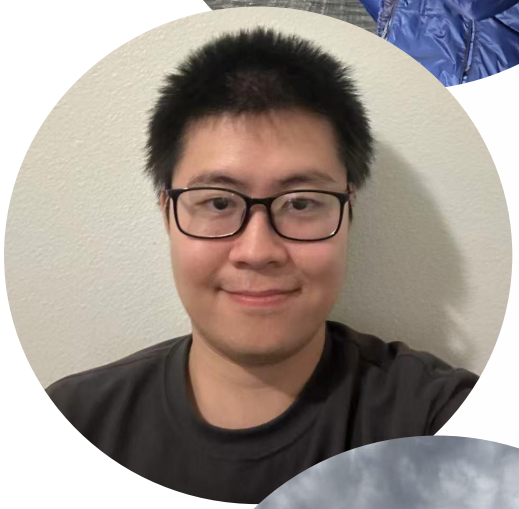
Key learnings

Across three interviews, we find that people enjoy physical activities most when they feel in control and immersed in the experience.

Despite differences in activity and skill level, all participants described moments of uncertainty, especially around form or technique. While they prefer figuring things out on their own, brief and well supports help restore confidence without breaking flow.

Next steps

In GR2, we will explore how support with AI can be offered only at moments of uncertainty while preserving autonomy and enjoyment. We will conduct additional interviews, refine POVs, and develop HMWs that focus on timing, subtlety, and user controls.



Thanks!

UC SANTA BARBARA